

CSCI 5922

Fundamentals of Neural Networks and Deep Learning

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Electives	Machine Learning Engineer (73.5 / 100)	62.9 / 100
SPECIALIZATION On-Campus Elective		

63	98	+47
CURRENT AVG SCORE across 6 job roles	POTENTIAL SCORE with recommended additions	MAX SCORE GAIN best single-role improvement

CURRENT COURSE DESCRIPTION

Covers fundamentals of neural networks and deep learning and their use to address many societal problems. Students will learn to design and implement neural network architectures, train them, and evaluate their performance. Included will be examination of popular architectures such as fully connected, convolutional, recurrent, and transformer-based networks alongside learning strategies such as backpropagation, regularization, and transfer learning. Students will also gain practical, hands-on experience by applying learned skills to analyze visual and textual data.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	73.5	100.0	+26.5	MEDIUM
ML Engineer	72.2	100.0	+27.8	MEDIUM
AI Engineer	67.8	100.0	+32.2	HIGH
Data Scientist	62.6	100.0	+37.4	HIGH
Applied Scientist	52.6	100.0	+47.4	HIGH
MLOps Engineer	48.5	85.8	+37.3	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 4 of 6 roles

systems

Business Communication 2 of 6 roles

insights

business

Deep Learning 2 of 6 roles

cnn

training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1 **HIGH** **Add System Design coverage** — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

- 2 **MEDIUM** **Consider Business Communication content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: insights, business.

- 3 **MEDIUM** **Consider Deep Learning content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: cnn, training.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

73.5 → 100.0 (+26.5 pts)

analytics tools exploratory business mobile development science models

ML Engineer

72.2 → 100.0 (+27.8 pts)

cnn discovery machine systems models build impact product

AI Engineer

67.8 → 100.0 (+32.2 pts)

perception models build technical systems technology detection machine

Data Scientist

62.6 → 100.0 (+37.4 pts)

product analytics mitigation insights social build approaches

Applied Scientist

52.6 → 100.0 (+47.4 pts)

discovery scientific research literature models technical systems industry

MLOps Engineer

48.5 → 85.8 (+37.3 pts)

detection text systems machine image training tools models

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥ 200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.